

Carewell Pharma - A Family of Learning

search CAREWELL PHARMA on YouTube & Watch Videos Syllabus wise

UNIT-II

- Pharmacology of drugs acting on Cardiovascular system

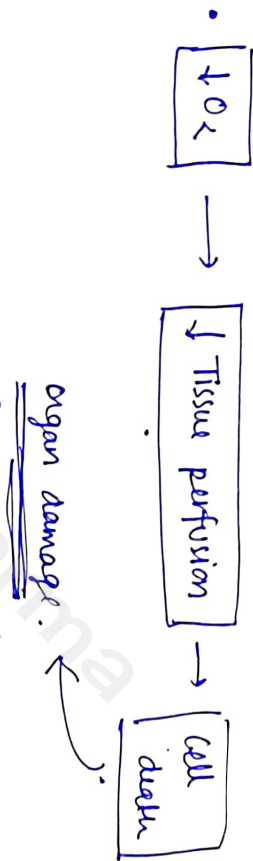
DRUGS USED IN THE THERAPY OF SHOCK

Shock :- It is a condition in which our body's cells does not get proper amount of Or supply (Hypoxia) which results in decreasement in ↓ tissue perfusion.

which leads to —

- Organ damage
- Heart failure
- Brain stroke
- ↑ Neurosis
- unconsciousness

and also may leads to death if save



• causes of shock :-

• ↓ in blood volume (blood supply)

- excessive loss of body fluids
- Haemorrhage (escape of blood from a rupture blood vessels)
- Trauma
- severe burns
- allergic reactions
- poisoning etc —

- Dehydration
- Diarrhoea

• Sign & symptoms of shock :-

- Nausea & Vomiting
- Pale skin (due to lack of blood).
- Shortness of breathing
- Rapid breathing
- Sweating
- Cold hand & feet

Carewell Pharma - A Family of Learning

search CAREWELL PHARMA on YouTube & Watch Videos Syllabus wise

• Types of shock →

There are mainly five types of shock:—

- 1) Hypovolemic shock
- 2) Cardiogenic shock
- 3) Anaphylactic shock
- 4) Septic shock
- 5) Neurogenic shock

① Hypovolemic shock →

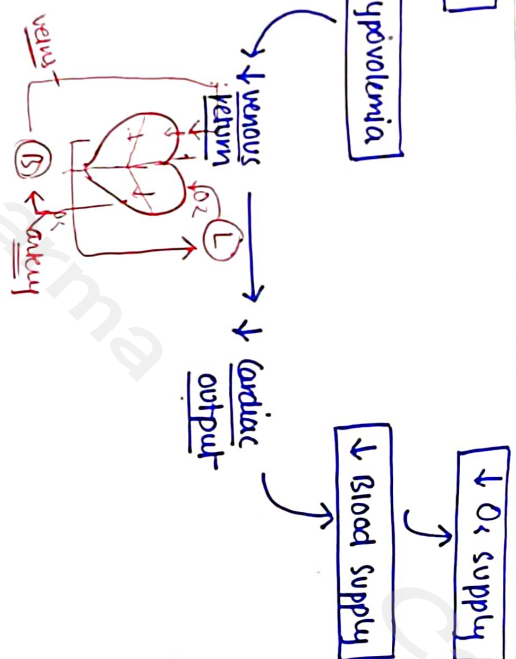
Hypo + volemic
↓ low volume } → low volume of blood
↓ ↓ O₂

• It is one of the type of shock which caused due to low volume of blood.

- Causes →
- ↑ Blood loss
 - Trauma
 - Haemorrhage
 - Electrolyte loss
 - Diarrhoea
 - Vomiting etc--

• Any causes

↓ Hypovolemia



• Treatment -

- Saline • Plasma • IV fluids (bloods)
- Plasma expands ^{↑ volume} - those agents which have relatively high molecular weight and boost the plasma volume by increasing the osmotic pressure.

② Cardiogenic shock →

It is one of the type of shock which is caused due to heart disease mainly due to myocardial infarction (MI) +

Carewell Pharma - A Family of Learning

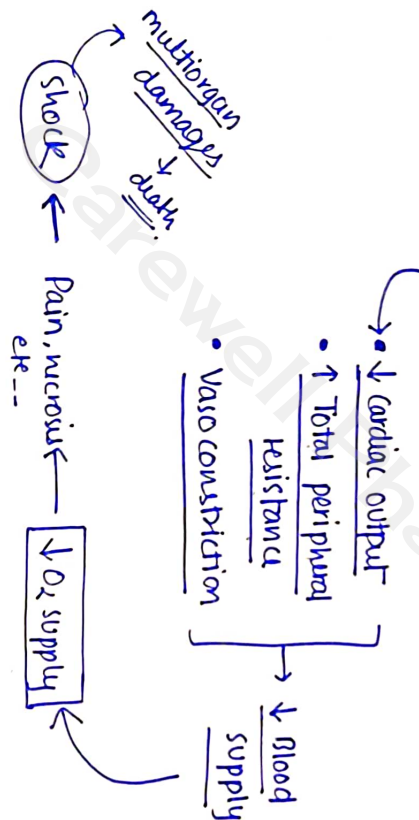
search CAREWELL PHARMA on YouTube & Watch Videos Syllabus wise

Angina pectoris [$\downarrow$ O₂ supply]

- causes →
- Myocardial infarction
 - Angina pectoris
 - Heart failure
 - Some other heart diseases
- ↓ O₂ supply
due to res in blood supply

Any cause

Cardiogenic shock



Treatment -

- Vasodilators (nifedipine)
- O₂ administration
- pain relievers
- Some drugs to treat CUS diseases

③ Anaphylactic shock →

It is one of the type of shock which is caused due to some allergic reactions such as hypersensitivity.

- causes →
- ↑ release of histamine and other mediators.

Treatment -

- Epinephrine → ↑ CO, + inotropic
- O₂ administration
- Corticosteroids → vasodilation, + inotropic

Anaphylactic shock



④ Septic shock →

It is due to some bacterial infection such as gram (+)ve and gram (-)ve bacteria.

- caused → ↑ endotoxins liberation.

Septic shock



Carewell Pharma - A Family of Learning

search CAREWELL PHARMA on YouTube & Watch Videos Syllabus wise

• Treatment -

- antibiotics therapy
- Or administration
- corticosteroids (large amount)

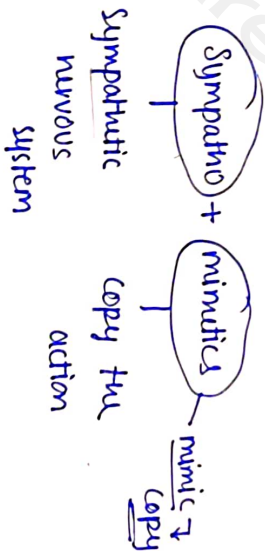
⑤ Neurogenic shock →

It is caused by damage to the nervous system.

- treatment - phenylephrine

DRUGS USED - classification

① Sympathomimetics agents →



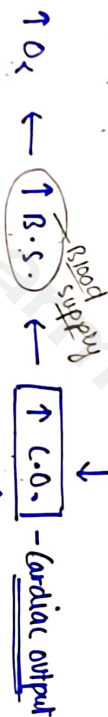
- those drugs which copy the action of Sympathetic Nervous system and give similar action.

i) Epinephrine →

- used to treat anaphylactic & septic shock.

- It binds on both α (alpha) and β (beta) receptor and cause -

- (+ve) inotropic effect → ↑ force of contraction



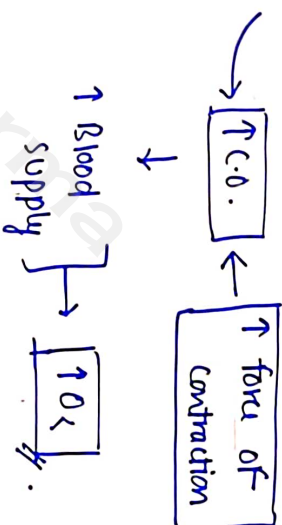
- β_2 receptor → Heart rate ↑

ii) Nor-epinephrine →

- used to treat cardiogenic & septic shock.

- β_1 Beta receptor, • (+ve) inotropic effect

- Vasodilation of coronary artery



Website :- www.carewellpharma.in

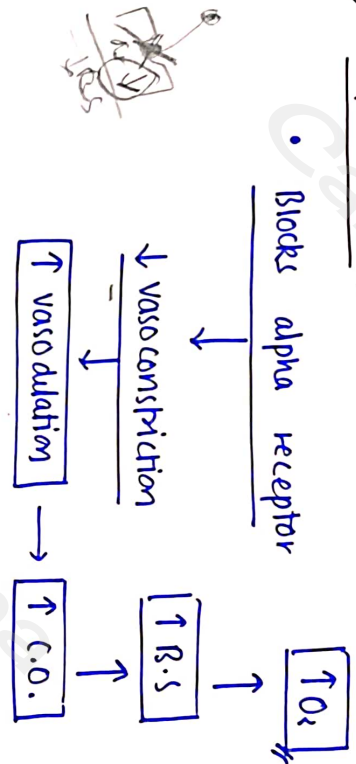
Visit website & Download free handwritten Notes

Carewell Pharma - A Family of Learning

search CAREWELL PHARMA on YouTube & Watch Videos Syllabus wise

② Alpha blockers →

- Blocks alpha receptor

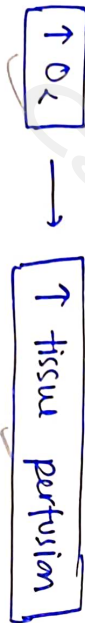


— used to treat cardiogenic shock.

eg. Prazosin, Phenoxybenzamine etc--

③ Oxygen (O₂) →

- Administration of O₂ can treat all kind of shock.



④ Plasma expanders →

- used to treat hypovolemic shock.

- They boost the plasma volume by increasing the osmotic pressure.

↑ Plasma volume expanders

↑ Plasma Volume

↑ Blood volume

↑ Blood supply

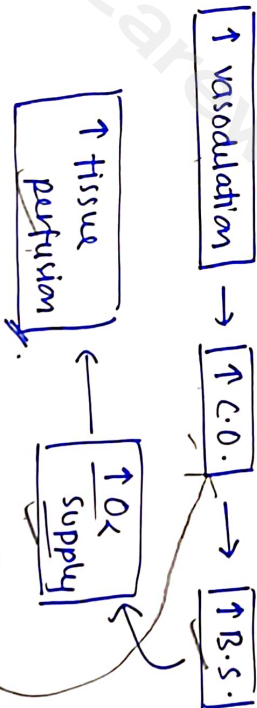
↑ O₂ supply

eg Dextran etc—

⑤ Corticosteroids →

— mainly used to treat anaphylactic and septic shock.

- They mainly give their action by increasing vasodilation



- They also have some (+ve) ionotropic effects

eg. Hydrocortisone, Methylprednisolone, Dexamethasone etc--